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HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Designing an ETL Pipeline for Analyzing Product Prices and Status from Lazada E-commerce Data

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2. Objectives and Proposed Solution
3. System Design
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1. Introduction

- Over time, product data on e-commerce platforms like Lazada has **grown significantly in volume**.
- Product information is distributed across various APIs and sources, with **inconsistent structures, missing fields, or frequent changes** — making the collection, processing, and analysis of data challenging.



1. Introduction

- It is necessary to build a system for extracting and managing product data:
 - Collecting, processing – integrating, and analyzing essential information such as price, inventory, and product status.
 - The system must also handle unexpected scenarios such as missing key SKU data, sudden price changes that distort analytical results, or unsynchronized updates between data collection intervals.





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2. Objectives and Proposed Solution

- A complete pipeline is built to collect, store, process, and analyze product data:
 - Product data is collected from Lazada's API, including information on prices, inventory, status, and product attributes.
 - The system leverages big data technologies to store data in both batch and real-time formats. The data is standardized and combined with SKU and technical information to create a comprehensive analytical dataset.
 - The results are integrated into dashboards for visualization and monitoring. In the future, the system can be extended to forecast price trends and product availability.

2. Objectives and Proposed Solution

- Visualize the entire process of product data collection, processing, and analysis using clear charts, dashboards, and illustrative statistics.
- Experiment with anomaly detection techniques (e.g., sudden price spikes, unexpected inventory drops) to ensure the system's accuracy and reliability over time.
- Deploy the system using Docker and virtualization technologies to ensure flexibility, scalability, and operational stability.



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3. System Design

Lambda Architecture

- Combines **batch** and **real-time** processing
- Ensures both **accuracy** and **low latency**
- Includes three layers:
 - Batch Layer – processes large data sets for accuracy
 - Speed Layer – handles real-time data for fast updates
 - Serving Layer – take results from 2 layers above for querying

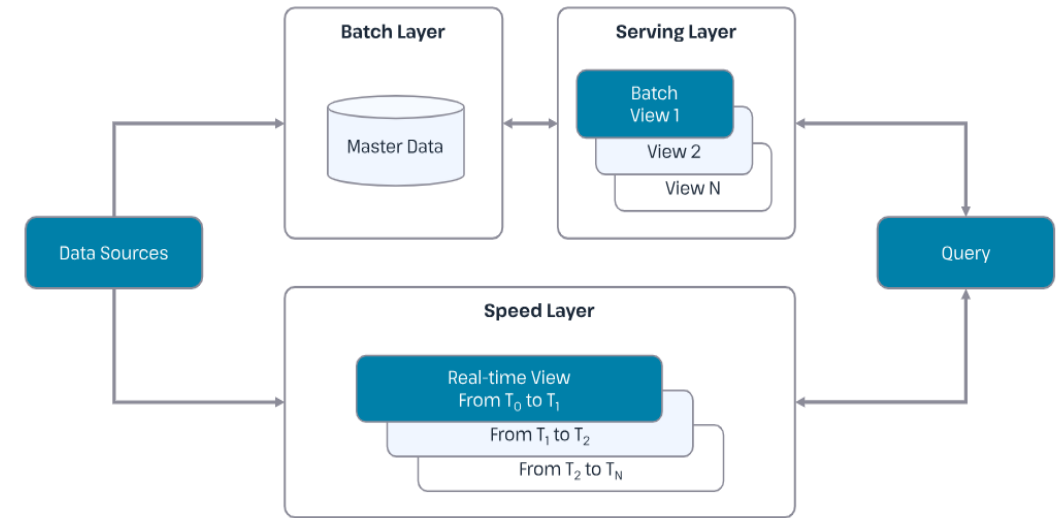


Figure 1: System architecture

3. System Design

System Overview

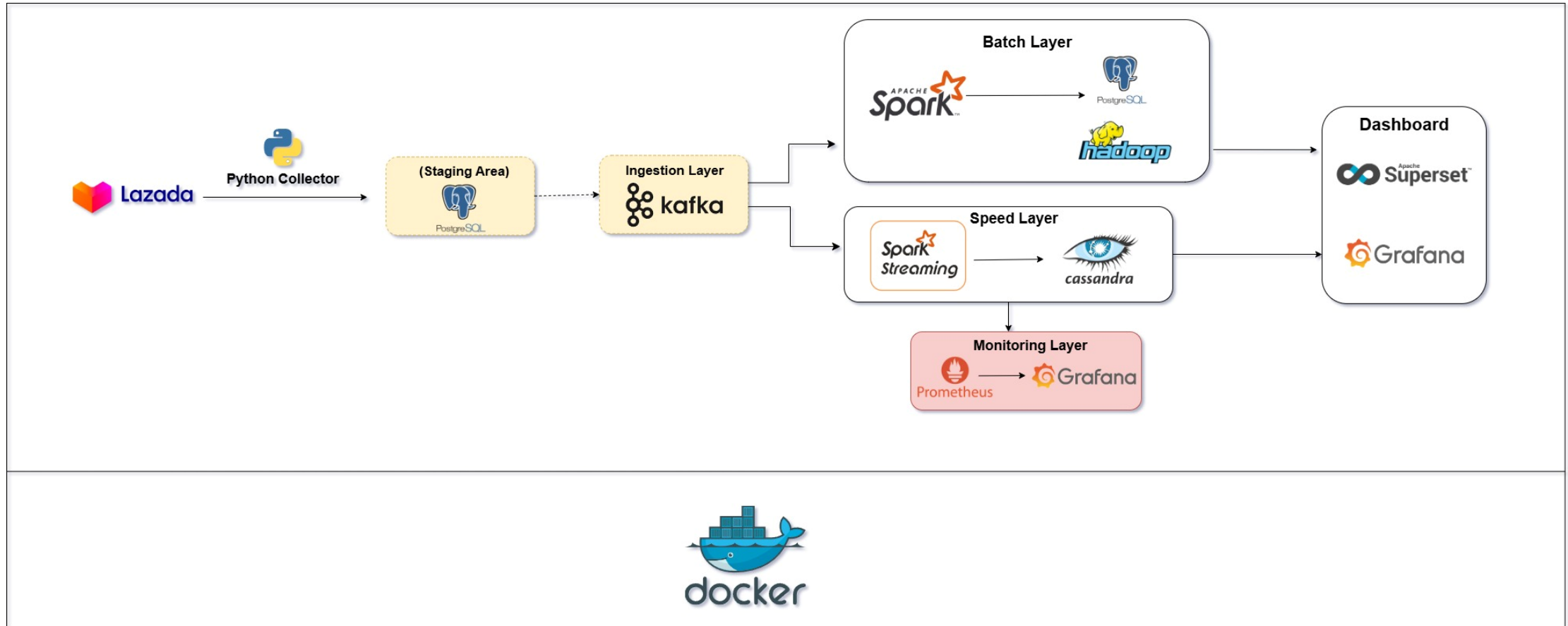


Figure 2: System Overview

3. System Design (Data Collection Module)

- **Purpose:** Collect product data from the Lazada platform
- **Components:**
 - **Lazada API** → Raw product data
 - **Python Collector:** configured as a Docker service, continuously calls the Lazada API using *last_updated* to fetch only new data.
 - **Kafka (Ingestion Layer):** Receives data and buffers messages for downstream processing
 - **PostgreSQL (Staging):** A copy of the raw data is also stored in a staging table for backup and audit purposes before processing

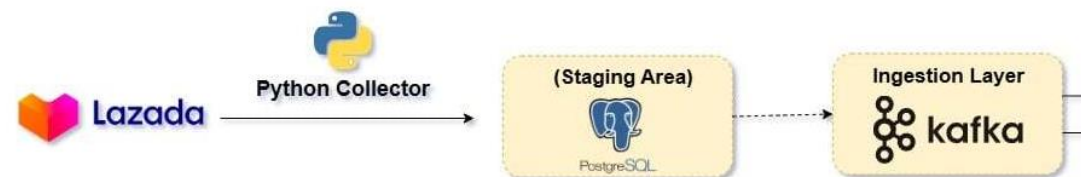


Figure 3: Data Collection Module from Lazada

3. System Design (Data Processing and Storage Module)

- **Purpose:** Transform and store product data for both real-time and batch analysis.
- **Components:**
 - **Spark Streaming Structure:** Reads from Kafka, processes and cleans data.
 - **PostgreSQL / Hadoop:** Store batch data for historical analysis.
 - **Cassandra:** Store real-time data for low-latency access.

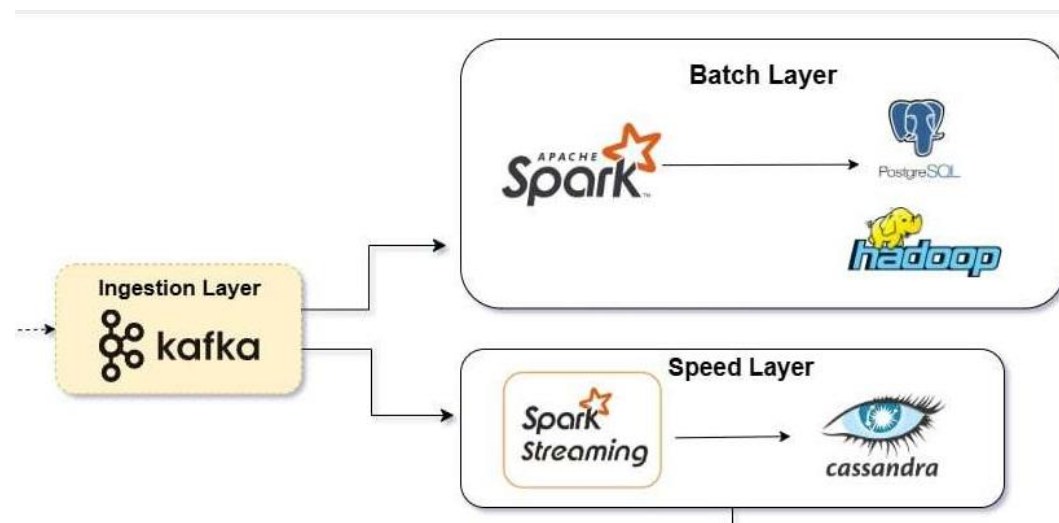


Figure 4: Data Processing and Storage Module

3. System Design (Data Visualization Module)

- **Purpose:** Display processed results through dashboards and reports
- **Components:**
 - **Apache Superset:** Connects to PostgreSQL to visualize:
 - Average price by category
 - Current stock levels of each SKU
 - Price fluctuation trends over time
 - **Grafana:** Connects to Cassandra to create real-time dashboards, visualizing up-to-date metrics such as product availability, live pricing,...

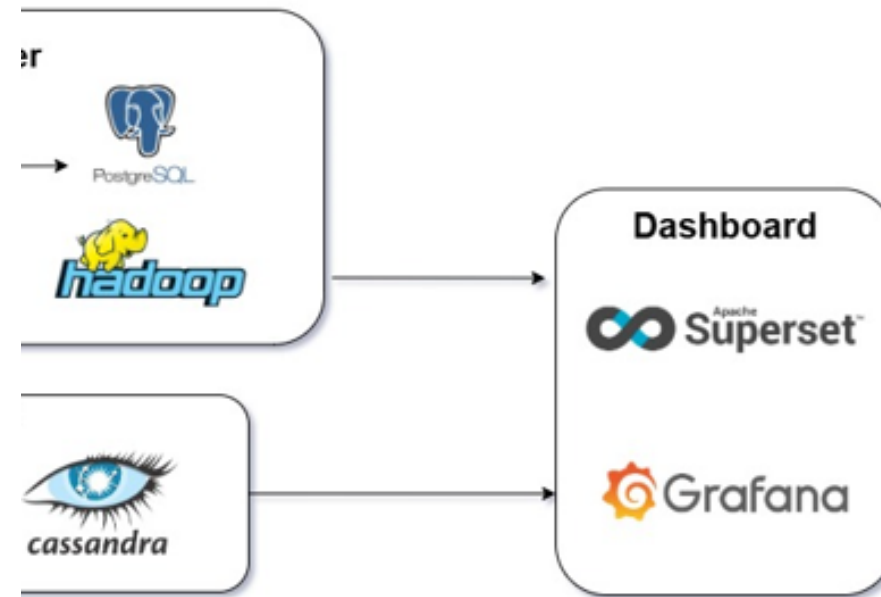


Figure 5: Data Visualization Module

3. System Design (Monitoring Module)

- **Purpose:** Monitor the performance and status of each system component
- **Components:**
 - **Prometheus:** Collects metrics from Spark, Kafka, and other services
 - **Grafana:** Visualizes system metrics (CPU, latency, message volume, etc.)

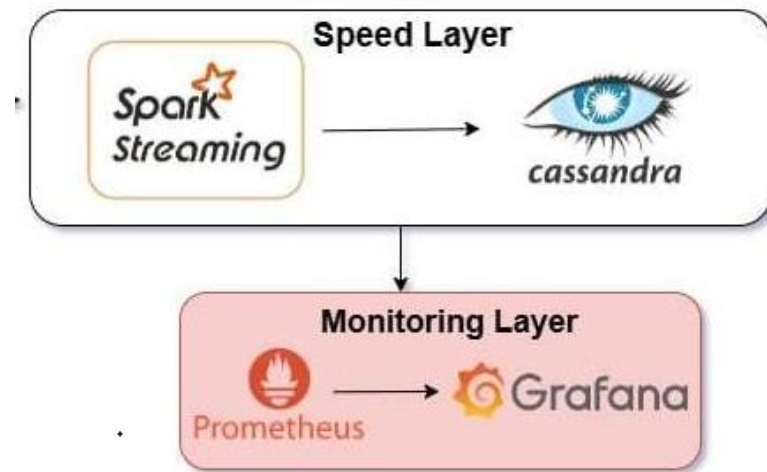


Figure 6: Monitoring Module



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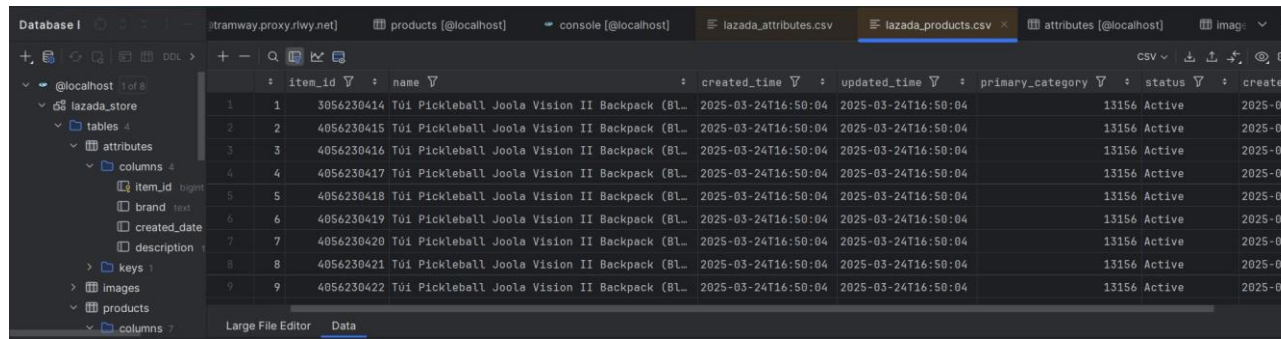
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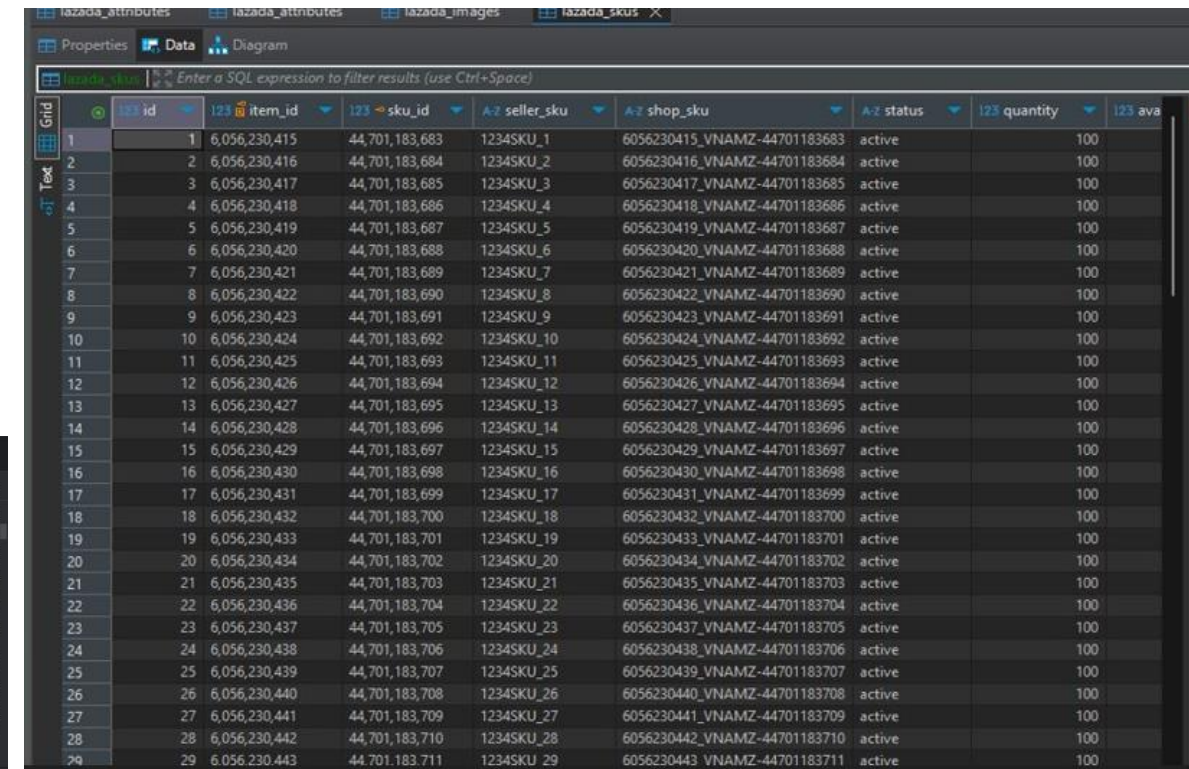
4. Experimental Result

Data Collection:

- The system successfully processed over 15M records, including data collected from the Lazada API and additional synthetic test data.
- Main dataset Includes two main tables:
 - Products: General info (name, category, status, timestamps)
 - SKUs: Detailed variants with seller/shop-specific codes



item_id	name	created_time	updated_time	primary_category	status	created
3056230414	Túi Pickleball Joola Vision II Backpack (BL...	2025-03-24T16:50:04	2025-03-24T16:50:04	13156	Active	2025-05
4056230415	Túi Pickleball Joola Vision II Backpack (BL...	2025-03-24T16:50:04	2025-03-24T16:50:04	13156	Active	2025-05
4056230416	Túi Pickleball Joola Vision II Backpack (BL...	2025-03-24T16:50:04	2025-03-24T16:50:04	13156	Active	2025-05
4056230417	Túi Pickleball Joola Vision II Backpack (BL...	2025-03-24T16:50:04	2025-03-24T16:50:04	13156	Active	2025-05
4056230418	Túi Pickleball Joola Vision II Backpack (BL...	2025-03-24T16:50:04	2025-03-24T16:50:04	13156	Active	2025-05
4056230419	Túi Pickleball Joola Vision II Backpack (BL...	2025-03-24T16:50:04	2025-03-24T16:50:04	13156	Active	2025-05
4056230420	Túi Pickleball Joola Vision II Backpack (BL...	2025-03-24T16:50:04	2025-03-24T16:50:04	13156	Active	2025-05
4056230421	Túi Pickleball Joola Vision II Backpack (BL...	2025-03-24T16:50:04	2025-03-24T16:50:04	13156	Active	2025-05
4056230422	Túi Pickleball Joola Vision II Backpack (BL...	2025-03-24T16:50:04	2025-03-24T16:50:04	13156	Active	2025-05



id	item_id	sku_id	seller_sku	shop_sku	status	quantity	ava
1	6,056,230,415	44,701,183,683	1234SKU_1	6056230415_VNAMZ-44701183683	active	100	
2	6,056,230,416	44,701,183,684	1234SKU_2	6056230416_VNAMZ-44701183684	active	100	
3	6,056,230,417	44,701,183,685	1234SKU_3	6056230417_VNAMZ-44701183685	active	100	
4	6,056,230,418	44,701,183,686	1234SKU_4	6056230418_VNAMZ-44701183686	active	100	
5	6,056,230,419	44,701,183,687	1234SKU_5	6056230419_VNAMZ-44701183687	active	100	
6	6,056,230,420	44,701,183,688	1234SKU_6	6056230420_VNAMZ-44701183688	active	100	
7	6,056,230,421	44,701,183,689	1234SKU_7	6056230421_VNAMZ-44701183689	active	100	
8	6,056,230,422	44,701,183,690	1234SKU_8	6056230422_VNAMZ-44701183690	active	100	
9	6,056,230,423	44,701,183,691	1234SKU_9	6056230423_VNAMZ-44701183691	active	100	
10	6,056,230,424	44,701,183,692	1234SKU_10	6056230424_VNAMZ-44701183692	active	100	
11	6,056,230,425	44,701,183,693	1234SKU_11	6056230425_VNAMZ-44701183693	active	100	
12	6,056,230,426	44,701,183,694	1234SKU_12	6056230426_VNAMZ-44701183694	active	100	
13	6,056,230,427	44,701,183,695	1234SKU_13	6056230427_VNAMZ-44701183695	active	100	
14	6,056,230,428	44,701,183,696	1234SKU_14	6056230428_VNAMZ-44701183696	active	100	
15	6,056,230,429	44,701,183,697	1234SKU_15	6056230429_VNAMZ-44701183697	active	100	
16	6,056,230,430	44,701,183,698	1234SKU_16	6056230430_VNAMZ-44701183698	active	100	
17	6,056,230,431	44,701,183,699	1234SKU_17	6056230431_VNAMZ-44701183699	active	100	
18	6,056,230,432	44,701,183,700	1234SKU_18	6056230432_VNAMZ-44701183700	active	100	
19	6,056,230,433	44,701,183,701	1234SKU_19	6056230433_VNAMZ-44701183701	active	100	
20	6,056,230,434	44,701,183,702	1234SKU_20	6056230434_VNAMZ-44701183702	active	100	
21	6,056,230,435	44,701,183,703	1234SKU_21	6056230435_VNAMZ-44701183703	active	100	
22	6,056,230,436	44,701,183,704	1234SKU_22	6056230436_VNAMZ-44701183704	active	100	
23	6,056,230,437	44,701,183,705	1234SKU_23	6056230437_VNAMZ-44701183705	active	100	
24	6,056,230,438	44,701,183,706	1234SKU_24	6056230438_VNAMZ-44701183706	active	100	
25	6,056,230,439	44,701,183,707	1234SKU_25	6056230439_VNAMZ-44701183707	active	100	
26	6,056,230,440	44,701,183,708	1234SKU_26	6056230440_VNAMZ-44701183708	active	100	
27	6,056,230,441	44,701,183,709	1234SKU_27	6056230441_VNAMZ-44701183709	active	100	
28	6,056,230,442	44,701,183,710	1234SKU_28	6056230442_VNAMZ-44701183710	active	100	
29	6,056,230,443	44,701,183,711	1234SKU_29	6056230443_VNAMZ-44701183711	active	100	

Figure 7: Data successfully loaded into database

4. Experimental Result

Product data is stored in HDFS:

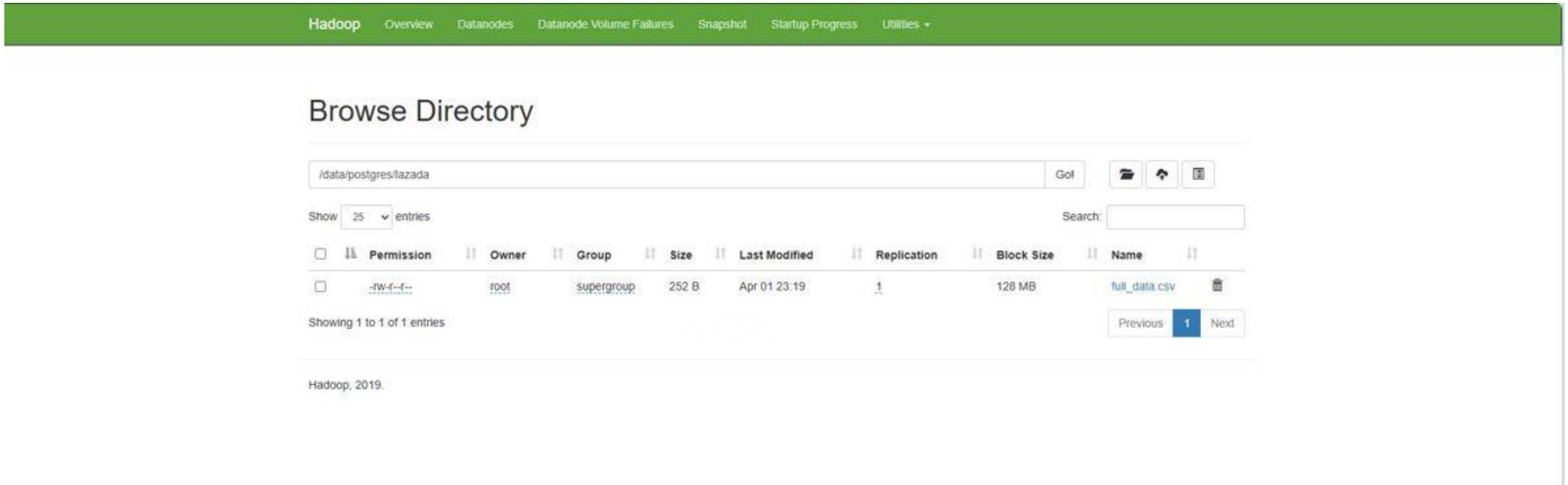


Figure 8: Hadoop user interface

4. Experimental Result

Data Visualization: Superset's dashboards for store overview, price analysis, and inventory analysis.

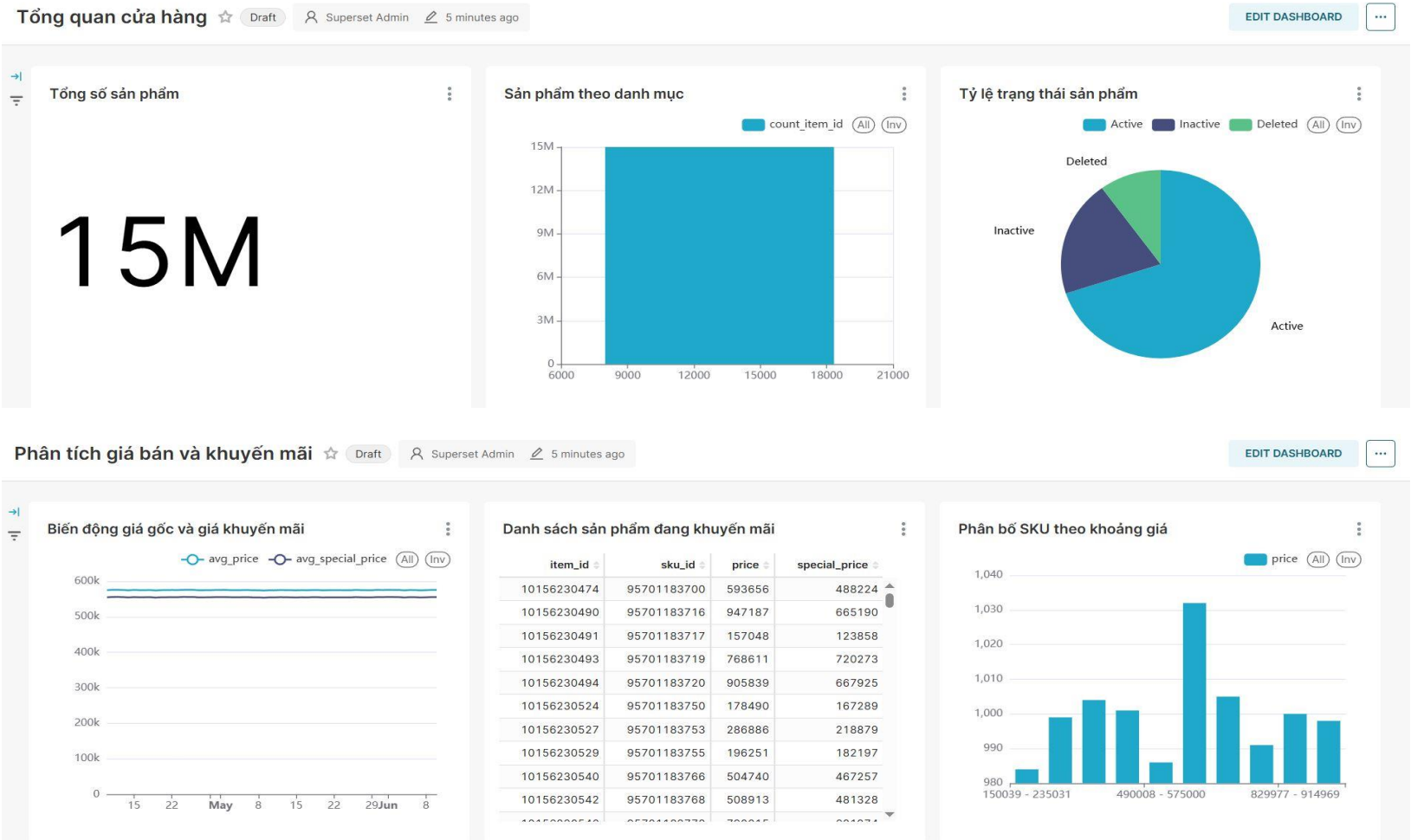


Figure 9: Store analysis charts

4. Experimental Result

Data Visualization: Superset's dashboards for store overview, price analysis, and inventory analysis.

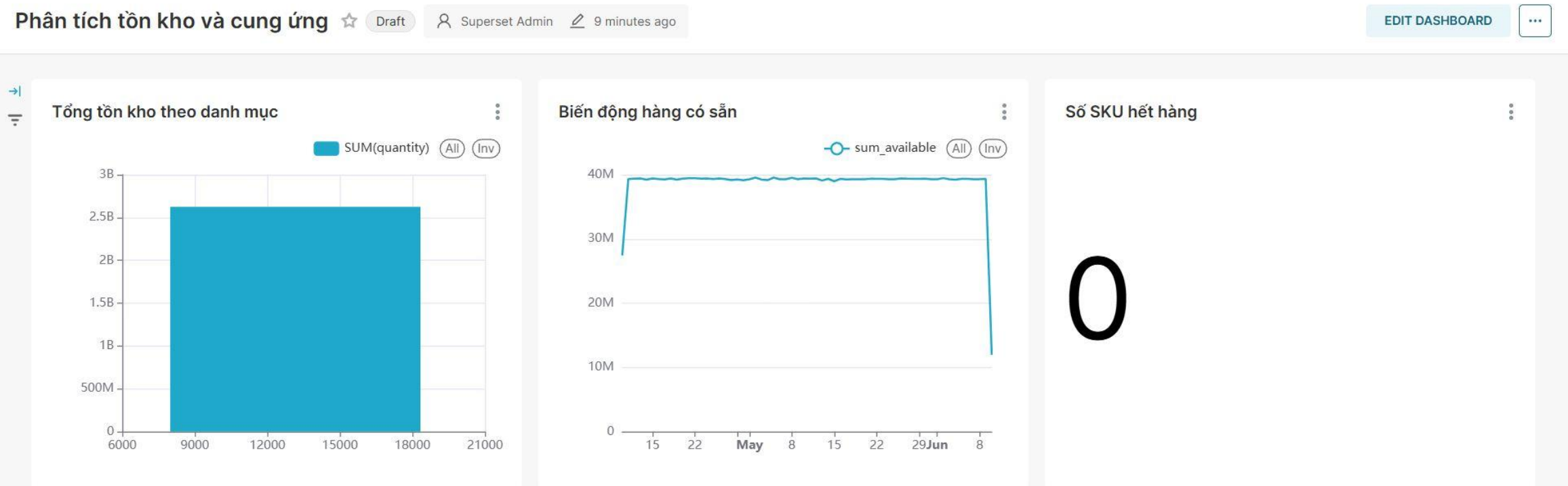


Figure 10: Store analysis charts

4. Experimental Result

Data Visualization: Grafana's realtime dashboard price analysis, and inventory analysis.

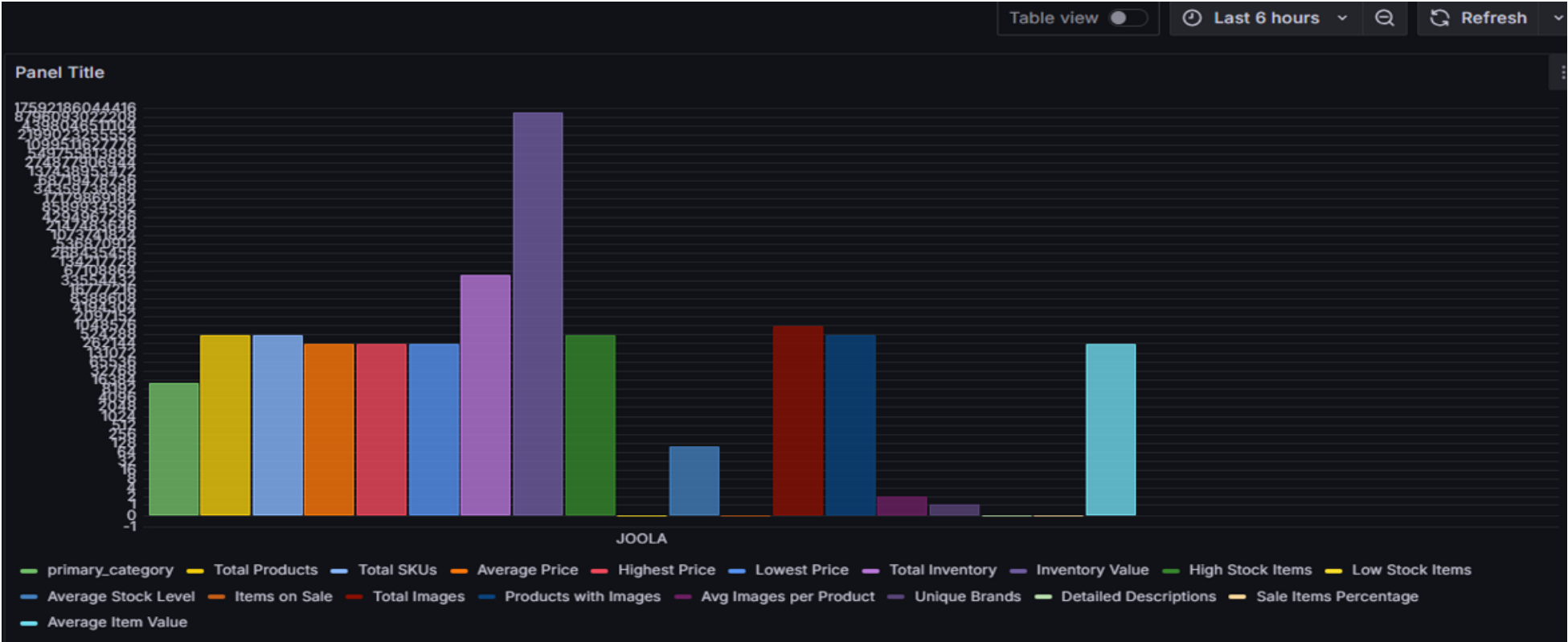


Figure 10: Store analysis charts

For example, this real-time bar chart visualizes JOOLA’s pricing, inventory, product, marketing, and promotion metrics from **Cassandra**—supporting strategic business decisions on Lazada.

4. Experimental Result

System Monitoring:

- Prometheus collected metrics from Kafka, Spark, and custom exporters.
- Grafana dashboards visualized real-time system status.

Alert Rules (Rule Health):

- **PriceVolatility**: triggers when product price fluctuates sharply
- **TechnicalIndicatorAlert**: detects anomalies in calculated indicators
- **ProductPriceSpike**: alerts on sudden price increases
- **LowInventory**: warns when product stock falls below threshold

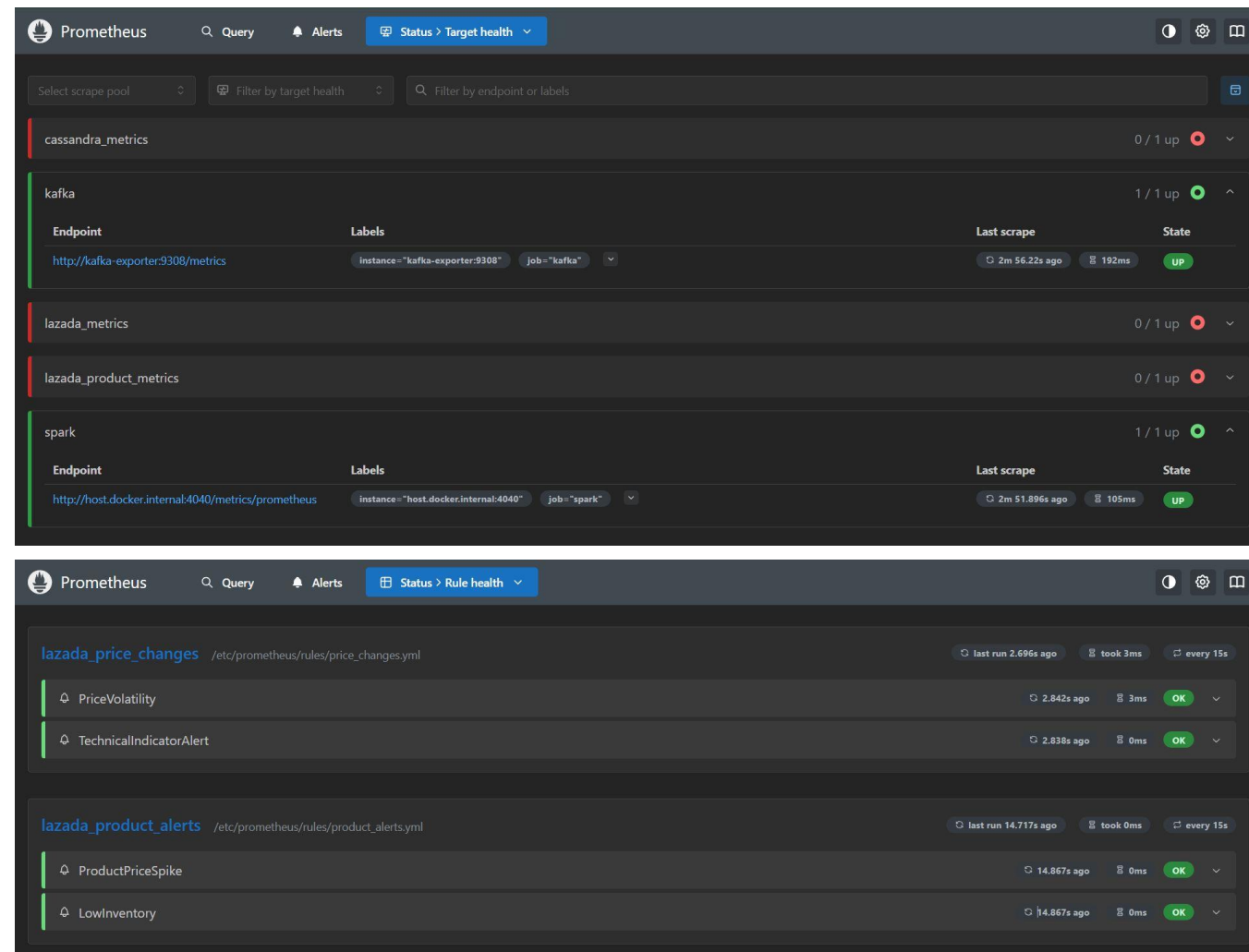


Figure 11: Prometheus monitoring targets and alert rules

4. Experimental Result

Spark Monitoring Dashboards:

- Displays GC count, GC time, and number of active jobs.
- Spark is running stably with no backlog; memory usage is efficient.

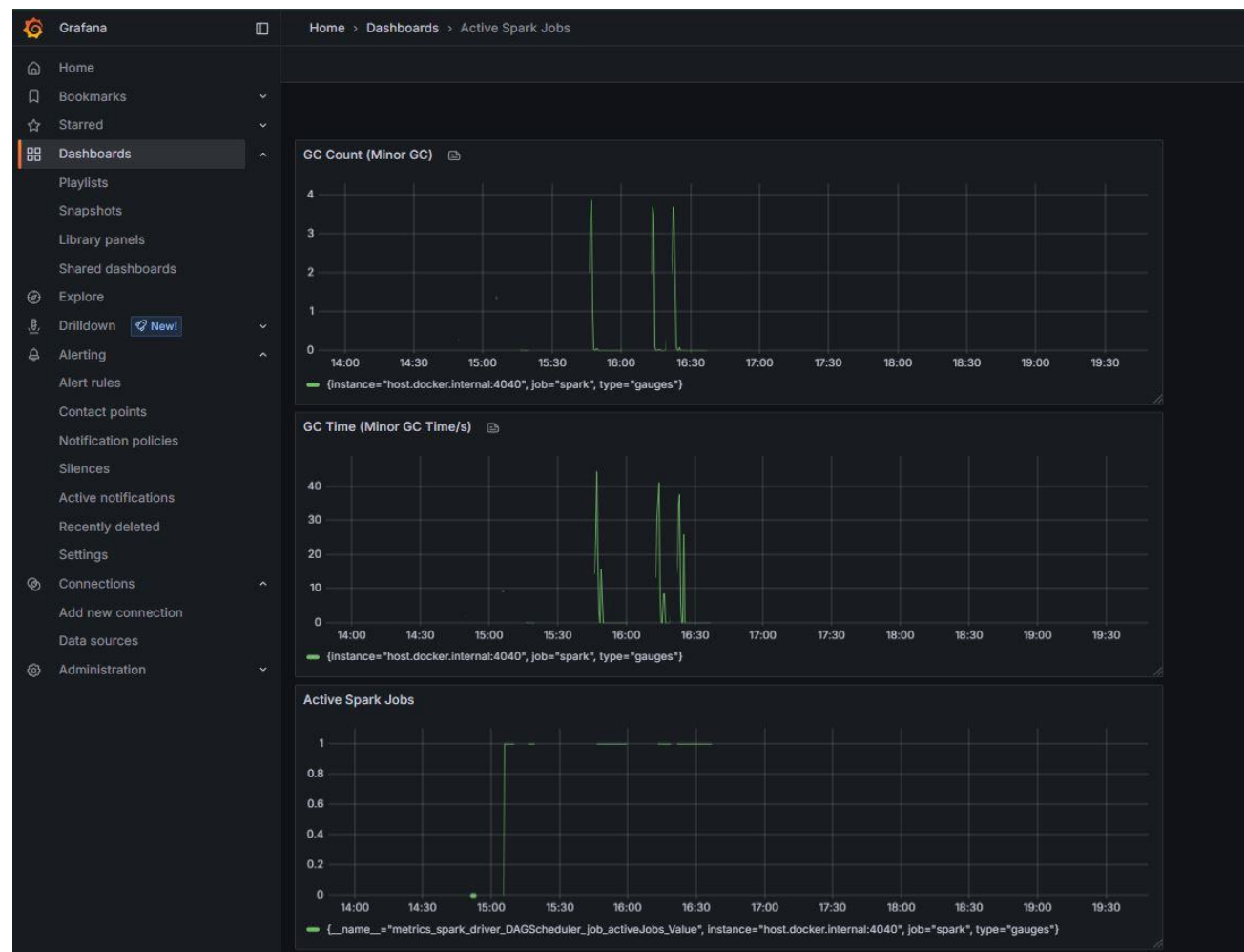


Figure 12: Spark monitoring dashboards

4. Experimental Result

Cassandra Monitoring Dashboards:

- Displays the scrape duration chart (the time Prometheus takes to collect data from Cassandra).
- The values fluctuate slightly and update regularly → indicating that Prometheus is collecting data consistently.



Figure 13: Cassandra monitoring dashboards

4. Experimental Result

Kafka Monitoring Dashboards:

- The dashboard displays offset counts by topic, especially for the two main topics: **lazada-products** and **lazada-skus**.
- **Red alerts** appear when offsets approach or reach the limit, indicating potential queue overflow or slow consumer processing.

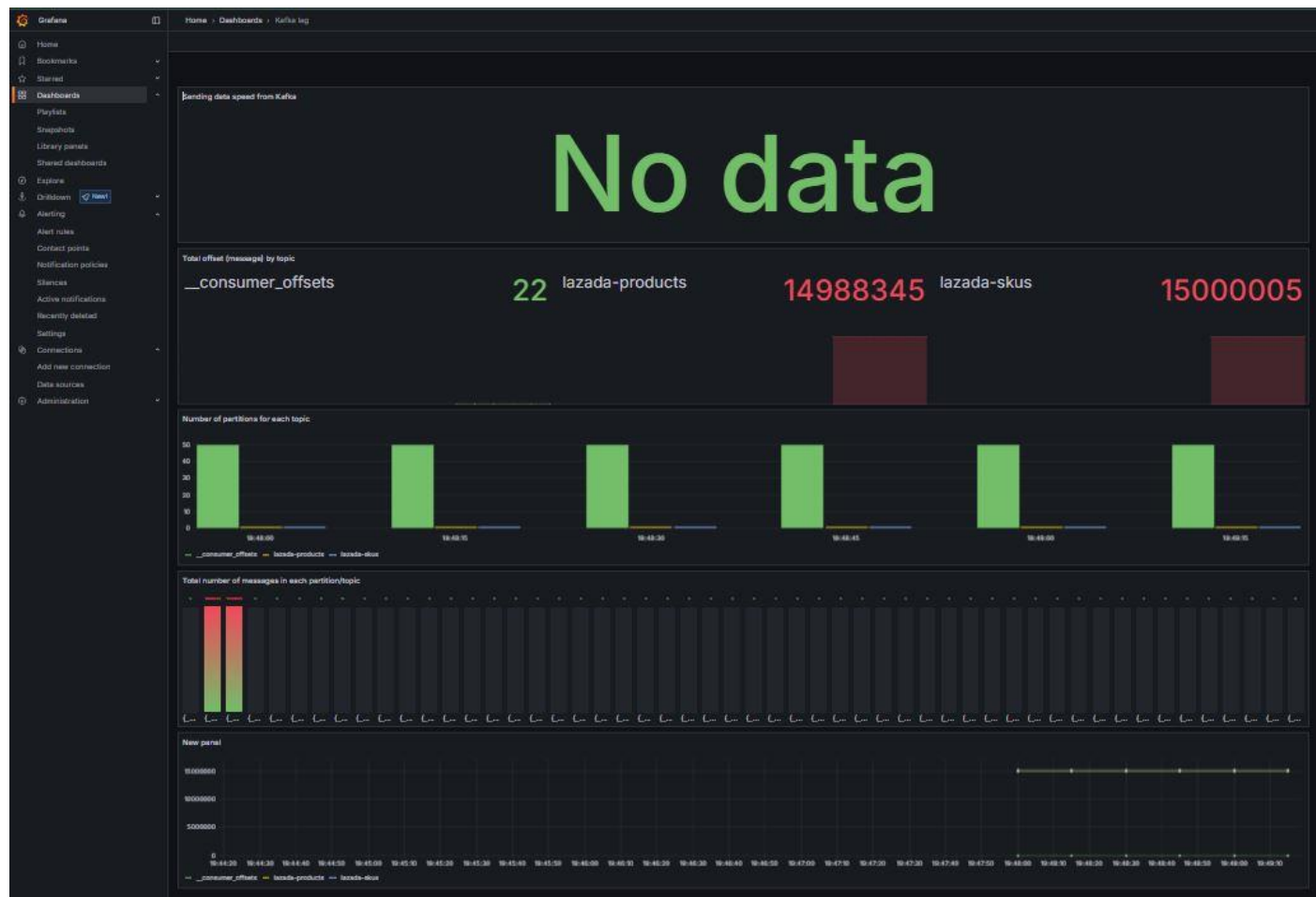


Figure 14: Kafka monitoring dashboards



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5. Conclusion

Key Achievements:

- Collected data from the Lazada Open API, with a complete pipeline for processing, cleaning, storage, and visualization.
- Delivered real-time analytics dashboards on categories, pricing, inventory, and performance.
- Ensured fault-tolerant architecture with scalable data integration.
- Applied modern tools: ETL pipelines, PostgreSQL, Grafana, and Prometheus.

Limitations:

- **Basic dashboards:** Visualizations are still simple and lack advanced KPIs and analytical depth.
- **Error handling:** The system needs improved retry logic, alerting mechanisms, and fault recovery strategies.

5. Conclusion

Future Development:

- **Data Expansion:**

- Upgrade API tier, add reviews/logistics/orders
- Implement caching & backup snapshots

- **Architecture & Scalability:**

- Move to microservices, deploy on cloud
- Use Kubernetes, add Redis & task queues

- **Advanced Analytics:**

- Add ML models (forecasting, anomaly detection)

- **Monitoring & Reliability:**

- Enable alert notifications (email/Telegram)

- **User Interface:**

- Build an intuitive web-based dashboard



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